Introduction

Quality database design is essential to ensure high-performance relative to access, data entry, and maintenance. A well-designed database can reduce data management costs, improve data analysis options, change disparate data into a valuable resource, and improve data consistency. A poorly designed database may lead to costly data management, loss of data, lack of data protection, and misinterpretation of data or its trends.

While there are many perspectives on the structure of a database design lifecycle, for the purpose of this lab, the database design lifecycle includes the following three stages:

- Conceptual Design – In the Conceptual Design phase, data requirements and user operations are analyzed and mapped to a high-level conceptual data model, for example, an Entity-Relationship Diagram (ERD).

- Logical Design – In the Logical Design phase, the high-level conceptual data model is further analyzed, normalized to reduce data redundancy, and mapped to a relation schema.

- Physical Design – In the Physical Design phase, a database management system (DBMS) is selected.

In the Conceptual Design phase, database designers work with clients to elicit data requirements and functional requirements that may be expressed in the form of diagrams, tables, documents, or conversations between the clients and the design team. The database designers then create conceptual maps that succinctly describe the objects to be stored in a database, along with their properties and relationships to each other. This high-level description will use notational conventions that convey the meanings of the users' requirements and begin to show the structure of the data. These high-level mappings will define entity types, attributes, relationships, keys, and integrity constraints.

Conceptual maps can be used for requirements verification and validation in the Logical Design phase prior to implementation in the Physical Design phase. This lab focuses on the Conceptual Design phase and uses notational conventions that are defined by the Entity-Relationship (ER) model. It is common for the phases to influence each other (e.g., sharing the Logical Design with a client may lead to changes in the Conceptual Design).

In this lab, you will assume the role of a database designer who is creating an Enrollment database for a client, which is the registrar's office at a university. Your team has met with the client to map requirements, and your task is to develop the initial ERD for Students and Courses.

Lab Overview

This lab has four parts, which should be completed in the order specified.

1. In the first part of the lab, you will create an ER diagram and add entities and attributes.
2. In the second part of the lab, you will add relationships to your ER diagram.
3. In the third part of the lab, you will model weak entities in your ER diagram.
4. In the fourth part of the lab, you will interpret and analyze an ER diagram.

Finally, if assigned by your instructor, you will complete a challenge exercise that allows you to use the skills you learned in the lab to conduct independent, unguided work – similar to what you will encounter in a real-world situation.

Learning Objectives

Upon completing this lab, you will be able to

1. Define the constructs in an ER diagram and their corresponding visual representation.
2. Represent entities, attributes, and relationships in an ER diagram.
3. Explain the meaning of entities, attributes, and relationships in ER diagrams.
4. Explain the meaning of the cardinality and participation constraints on a relationship in an ER diagram.

5. Comprehend the requirements of a real-world database design expressed as a diagram.

Topology

This lab contains the following virtual machines. Please refer to the network topology diagram below.

- vWorkstation (Windows: Server 2022)

Tools and Software

The following software and/or utilities are required to complete this lab. Students are encouraged to explore the Internet to learn more about the products and tools used in this lab.

- Dia

Deliverables

Upon completion of this lab, you are required to provide the following deliverables to your instructor:

Hands-On Demonstration

1. Lab Report file including screen captures of the following:

- ER diagram with the Student entity and attributes
- ER diagram with the EnrollsIn relationship, Student and Course entities, and attributes for all three
- the entire ERD

2. Any additional information as directed by the lab:

- None

Challenge and Analysis

1. Lab Report file including screen captures of the following:

- Your catering application ER diagram final result in the Dia Canvas

2. Any additional information as directed by the lab:

- Document the names of all of the strong entities and their primary key attributes in the diagram
- Document the names of all of the binary relationships and their cardinalities in the diagram
- Document the names of the ternary relationships in the diagram
- Document the names of entities and relationships where the entity has total participation in the relationship
- Document the names of the weak entity, its identifying relationship, its owner entity, and its weak key
- Document the names of multi-valued attributes and the entity or relationship they belong to
- Document the names of composite attributes and the entity or relationship they belong to
- Document which attribute or attributes uniquely identifies the entities (1) Staff, (2) Corporation,

(3) Event, and (4) Venue

- Document all of the ER diagram elements associated with the weak entity Offering
- Document the combination of attributes needed to uniquely identify an instance of Offering (its weak key and its owner entity's primary key)
- Document the range (minimum and maximum) of instances of Staff that manage an Event using the cardinality of the Manages relationship and the participation indicated for Staff in Manages
- Document the range (minimum and maximum) of instances of Offering related to an Event using the cardinality of the Schedules relationship and the participation indicated for Offering in Schedules
- Document the range (minimum and maximum) of instances of Corporation related to Staff Event using the cardinality of the Caters relationship and the participation indicated for Corporation in Caters
- Document whether a value is associated with each attribute in the composite attribute shown in the diagram
- Document whether an instance of Staff can have 0 or more values for (1) roles, (2) title, and (3) sID
- Document what the dashed oval means for the attribute savings of the Manages relationship

Hands-On Demonstration

Note: In this section of the lab, you will follow a step-by-step walk through of the objectives for this lab to produce the expected deliverables.

1. **Review** the **Tutorial**.

   Frequently performed tasks, such as making screen captures and downloading your Lab Report, are explained in the Cloud Lab Tutorial. The Cloud Lab Tutorial is available from the User menu in the upper-right corner of the Student Dashboard. You should review these tasks before starting the lab.

2. **Proceed** with **Part 1**.

Part 1: Represent Entities and Attributes in an ER diagram

Note: Creating an ERD is one of the first steps in a typical database design process. The ERD allows a designer to raise questions for the client to clarify and refine the requirements before proceeding to the Physical Design and Implementation phases. Identifying issues such as missing or inconsistent information is important to do early in the design process when fixing the issues is easier and more cost effective compared to fixing them later in the process.

The ER model allows a database designer to express the conceptual properties of a database without requiring any specific details about the implementation platform used to create, populate, and retrieve information from the database. Those details are determined in the Physical Design phase. The ERD is an abstract representation of the data describing the real-world scenario that is important for the application.

An ERD is created by interpreting the user's requirements and expressing them in the notational conventions of the ER model (i.e., drawing a picture, where the notation used in the picture conveys specific information about the data). The entities that correspond to real-world objects or concepts can be expressed through ER model notation:

- *Entities* are represented by rectangles. An example of an entity is a student.
- The *attributes* of an entity are represented by ovals. An example of an attribute for a student entity is the student's name.
- *Relationships* between entities are represented by diamonds. An example of a relationship is a student (entity) *enrolling* in a course (entity).

In this part of the lab, you will create an entity with the following kinds of attributes: single-valued, primary keys, multi-valued, composite, and derived. Each of these will be defined and illustrated throughout the lab.

In the next steps, you will open the Dia application, which you will use to create the Enrollment ERD.

1. On the vWorkstation desktop, **double-click** the **Dia icon** to open the Dia application.

Dia Icon

2. **Maximize** the **Dia application window**.

Note: You should see the Flowchart view of the Dia graphical user interface. The main portion of the application window contains the Canvas, in which you will create diagrams. The Toolbox is situated in the left pane and contains controls and stencils that you will use to compose the elements of the diagrams on the Canvas.

The top 16 icons in the Toolbox include control icons:

- Modify Control (arrow)
- Textedit Control (vertical cursor)
- Magnify Control (magnifying glass)
- Scroll Control (crosshairs)

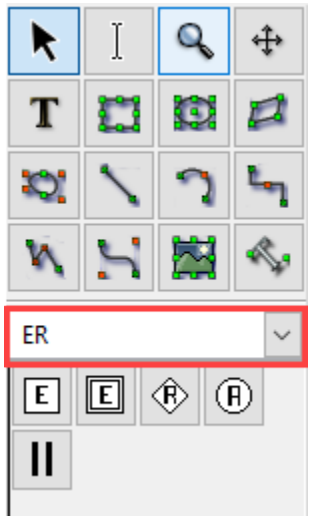
and the Basic Objects icons:

- Text
- Box
- Ellipse
- Polygon
- Beziergon
- Line
- Arc
- Zigzagline
- Polyline
- Bezierline
- Image
- Outline

The Special Objects drop-down list affects which Special Objects' icons are displayed below it. Its initial value is Flowchart; correspondingly, the icons below it represent stencils that are commonly used in creating Flowchart diagrams. You can read more about the Dia Canvas and the Toolbox in Dia's online documentation.

In the next step, you will load the Special Objects icons for an ERD.

3. In the Toolbox, **click** the **Special Objects drop-down list** and confirm that **Other sheets > ER is selected** to reveal the Special Objects icons for an Entity Relationship diagram.



Select the ER toolbox

Note: You should see that the icons below the Special Objects drop-down list have changed. The lab will refer to the Special Objects icons for the entity relationship stencils as the *ER Toolbox*. These icons represent stencils that are commonly used in creating Entity-Relationship diagrams:

- Entity – The rectangle button
- Entity with a weak key – The double-bordered rectangle button
- Relationship – The diamond button
- Attribute – The oval button
- Participation – The vertical line button

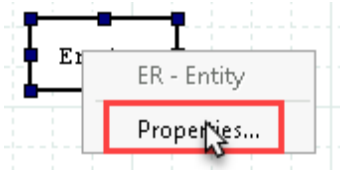
In the next steps, you will begin to create the Enrollment ERD by adding a Student entity.

4. In the ER Toolbox, **click** the **Entity button** and then **click a location near the center of the top of the Canvas** to add an entity stencil to your ERD.

Entity Button

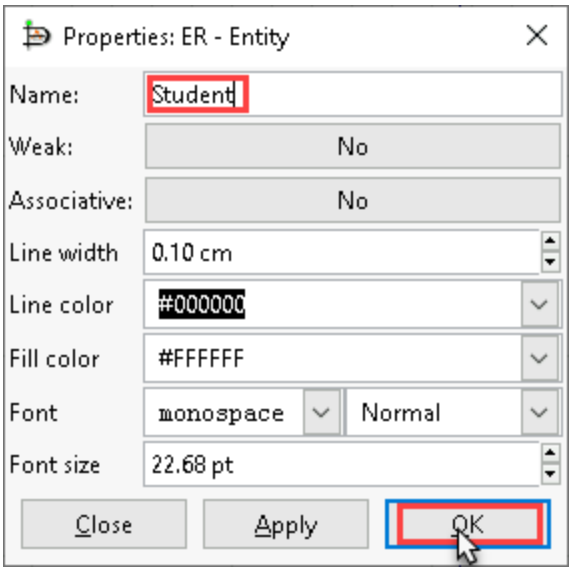
Note: You should see a rectangular stencil with the word “Entity” on the Canvas.

5. In the Canvas, **right-click** the **Entity rectangle** and then **select Properties** from the context menu to open the Properties: ER – Entity dialog box.



Open Entity Properties Dialog

6. In the Properties: ER – Entity dialog box, **type Student** in the Name: field; then **click** the **OK button** to rename the entity to *Student*.



Rename the Entity

Note: The features of an entity are represented by *attributes* in an ER diagram. The simplest form of an attribute is a feature that has a single value. This is known as a single-valued attribute. In the development stage of database development, the attribute will correspond to a column (field) with a specified datatype in a database table. Converting ERDs to database tables is outside the scope of this lab. In the conceptual design, it is important to show when an attribute is single valued.

The university keeps track of the total number of credits earned by each student. This can be represented by an attribute in the ERD. Because a student has only one value for the total amount of credits earned, the attribute should be single-valued. A single-valued attribute is represented by an oval with a single outline. In the naming convention for ERDs, it is customary to capitalize the names

of entities and to not capitalize the names of attributes.

Over the course of the next steps, you will add a credits attribute to the Student entity in the Enrollment ERD.

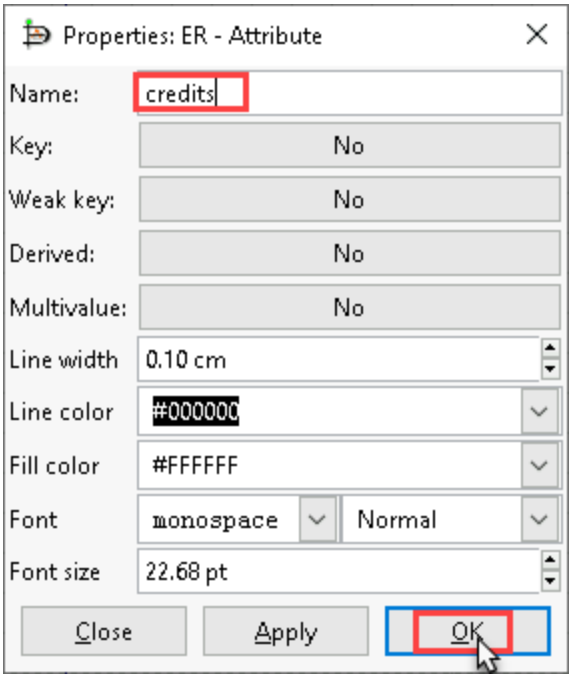
7. In the Toolbox, **click** the **Attribute button** and then **click a location to the left of the Student Entity in the Canvas** to add an attribute stencil to your ERD.

Attribute Button

Note: You should see an oval stencil with the word “Attribute” on the Canvas.

8. In the Canvas, **right-click** the **Attribute oval** and then **select Properties** from the context menu to open the Properties: ER – Attribute dialog box.

9. In the Properties: ER – Attribute dialog box, **type credits** in the Name: field; then **click** the **OK button** to rename the attribute to *credits*.



Change the Attribute Name

Note: In an ERD, the stencils of an entity and its attribute are connected by a solid line called a *connecting line*. Two entities may be connected by a *Participation line*. Dia uses the same icon to access a line stencil.

In the next steps, you will use the Participation icon to draw a connecting line to indicate that the *credits* attribute is a feature of the *Student* entity in your ERD.

10. In the ER Toolbox, **click** the **Participation button** and then **click** the **credits attribute** in the drawing pane to add a connecting line that originates with the credit attribute.

Participation Button

11. In the Canvas, **click and drag** the **green end of the connecting line to the Student entity** to complete the connection between the credits attribute and the Student entity.

Note: When a connecting line is ready to attach to a stencil on Dia's Canvas, the stencil's outline will briefly be highlighted in red. When you release the mouse, the connecting line's green endpoint will change to red. If the connecting line's endpoint remains green or changes to yellow, it has not attached to the stencil. If you wish to delete a connector, click on it and press the DEL (delete) key. The orange boxes on the line are segment connectors and can be moved to alter the layout of the lines.

Each entity should also have one or more attributes that allow it to be uniquely identified. It may be tempting to use an attribute like a name as the identifier for a student. However, it is possible for students to have the same name. Therefore, the university assigns an ID to each student. You will add an identifying attribute to the Student entity. When converting an ERD to a database table, an Entity's identifying attribute(s) will constitute the *primary key* of the database table that represents the Entity. A primary key is unique and must have a non-null value. Primary keys are designated in an ERD by underlining the name of the attribute. In Dia, this is managed through the Properties of the Attribute stencil.

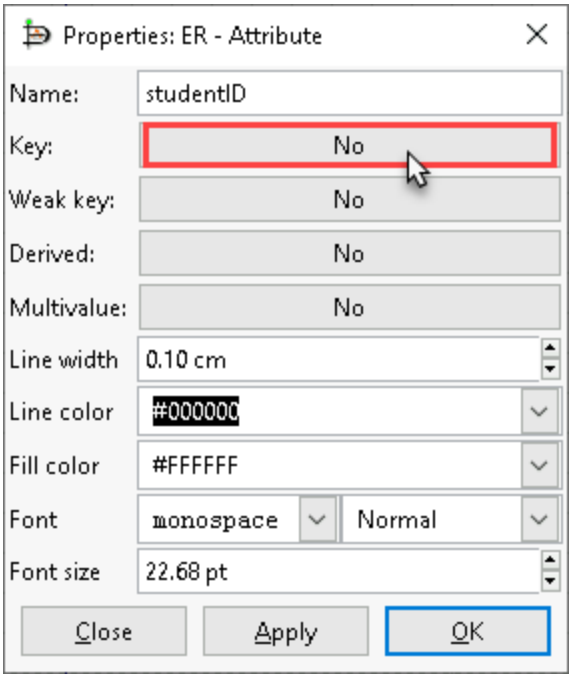
In the next steps, you will add a Student ID attribute, which will be used as an identifier for the Student entity.

12. In the ER Toolbox, **click** the **Attribute button** and then **click a location below the credits attribute in the Canvas** to add an attribute stencil to your ERD.

13. In the Canvas, **right-click** the **Attribute oval** and then **select Properties** from the context menu to open the Properties: ER – Attribute dialog box.

14. In the Properties: ER – Attribute dialog box, **type studentID** in the Name: field to rename the attribute to *studentID*.

15. In the Properties: ER – Attribute dialog box, **click** the **No button** of the Key: field to toggle its value to Yes; then **click** the **OK button** to close the Properties: ER – Attribute dialog box.



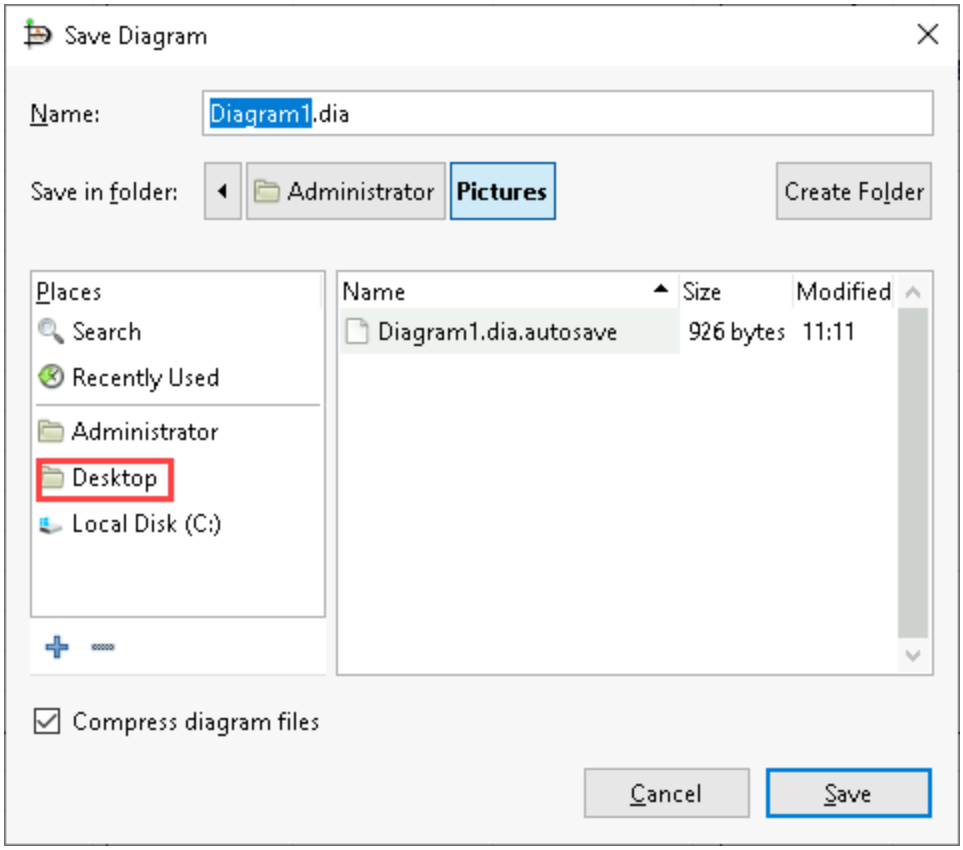
Key Field Button

Note: You should see that studentID is underlined inside its oval stencil. This graphically indicates that the studentID attribute is a key.

16. **Repeat steps 10–11** to connect the **studentID attribute** to the **Student entity**.

17. In the Dia menu bar, **select File > Save** to open the Save Diagram dialog box.

18. In the Save Diagram dialog box's Places pane, **select** the **Desktop folder** to change the Save in folder: location to Administrator > Desktop.



Desktop Folder in Places

19. In the Save Diagram dialog box, **replace the text in the Name: input field** with **EnrollmentERD.dia** and then **click** the **Save button** to save your diagram into a file called EnrollmentERD.dia on the vWorkstation Desktop.

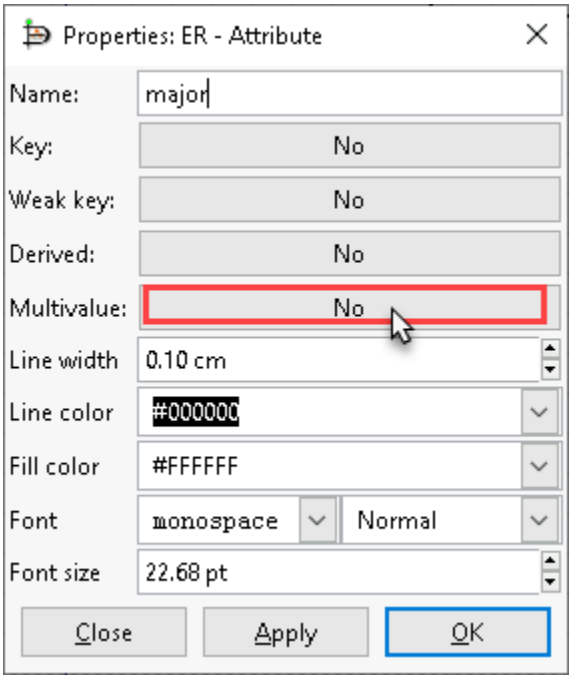
Note: If at any time you need to pause the lab, you can double-click the EnrollmentERD.dia diagram icon on the Desktop to continue editing the saved ERD. You will need to maximize the Dia application window again. Upon re-opening the file, you may need to use the scroll bars to locate your stencils in the Canvas.

Some attributes can have a set of values rather than just one value. For example, if a student can have multiple majors, this can be represented in an ER diagram as a multi-valued attribute. Examples of the values could be the set {CS} or the set {IS, MTH}. The set's value can also be the empty set (Æ or {}). Multi-valued attributes are represented by a double oval in an ER diagram.

In the next steps, you will add a multi-valued attribute called major.

20. **Repeat steps 12–14** to create a new attribute with the value **major** in the Name: field.

21. In the Properties: ER – Attribute dialog box, **click** the **No button** for the Multivalue field to toggle its value to Yes; then **click** the **OK button** to close the Properties: ER – Attribute dialog box.



Multivalue Field Button

Note: You should see that the major attribute's stencil has a double-edged outline, indicating that it is a multi-valued attribute.

22. **Repeat steps 10–11** to **connect** the **major attribute** to the **Student entity**.

Note: Some attributes have subparts. An example is an address, which may be composed of a number, street, city, country, and postal code. A student's name can be represented by a first and last

name. In an ER diagram, an attribute with subparts is called a *composite attribute*. A composite attribute has two levels. The first is represented by an oval connected to the entity, and no value is associated with this level. For a student's name, the first level will be called name.

23. In the ER Toolbox, **click** the **Attribute button** and then **click a location below the Student Entity in the Canvas** to add an attribute stencil to your ERD.

24. In the Canvas, **right-click** the **Attribute oval** and then **select Properties** from the context menu to open the Properties: ER – Attribute dialog box.

25. In the Properties: ER – Attribute dialog box, **type name** in the Name: field to rename the attribute to *name*.

26. **Repeat steps 10–11** to **connect** the **name attribute** to the **Student entity**.

Note: The second level has attributes connected to the first level, and these attributes will have values. The second level of our student name attribute will have two attributes: firstName and lastName.

27. In the ER Toolbox, **click** the **Attribute button** and then **click a location below the name attribute in the Canvas** to add an attribute stencil to your ERD.

28. In the Canvas, **right-click** the new **Attribute oval** and then **select Properties** from the context menu to open the Properties: ER – Attribute dialog box.

29. In the Properties: ER – Attribute dialog box, **type firstName** in the Name: field to rename the attribute to *firstName*; then **click** the **OK button** to close the Properties: ER – Attribute dialog box.

30. In the ER Toolbox, **click** the **Participation button** and then **click** the **firstName attribute** in the drawing pane to add a connecting line that originates with the firstName attribute.

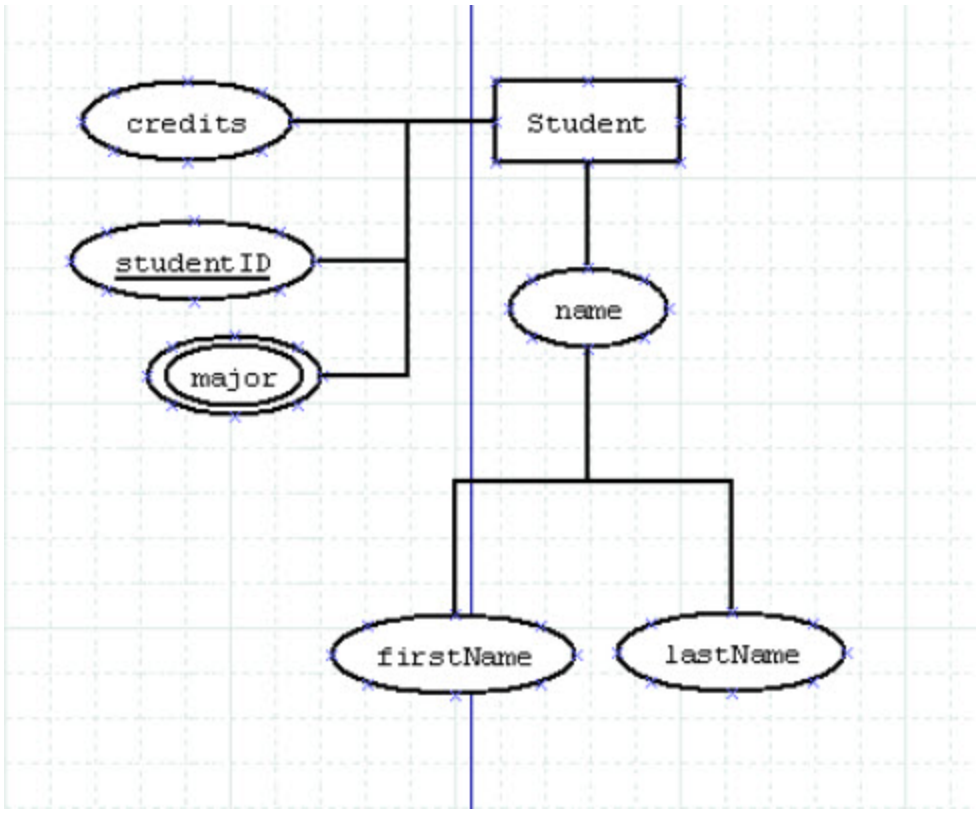
31. In the Canvas, **click and drag** the **green end of the connecting line to the name**

attribute to complete the connection between the firstName attribute (Level 2) and the name attribute (Level 1).

32. **Repeat steps 27–31** to create a new second-level attribute with the following features.

    - Canvas location: *to the right of the firstName attribute*
    - Name: **lastName**
    - Connecting line to: **name attribute stencil**

Note: Your ER diagram should have all of the same shapes and labels as this one, but your layout may not match exactly:



Example ER Diagram

In the next steps, you will add a derived or computed attribute. An example would be if a student's age was computed based on his or her birth date and the current date. The ER model does not include a way to specify the method for deriving the value, only that the value is derived. The method should be given as additional documentation to accompany the ER diagram. A derived attribute is

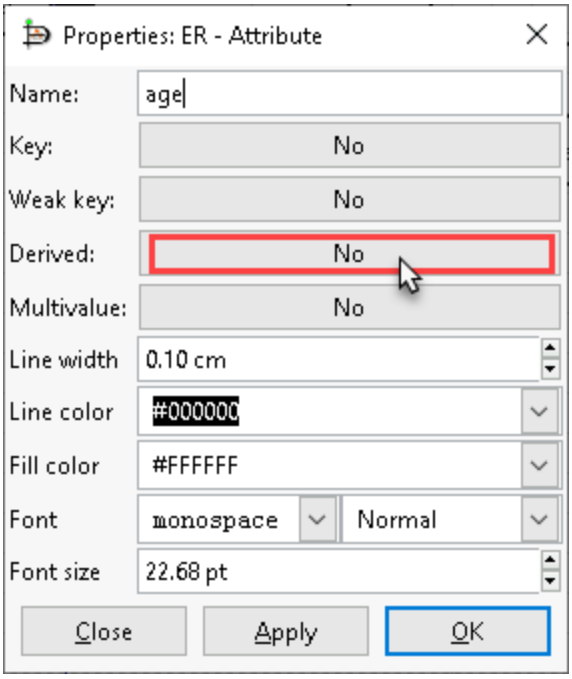
represented as a dashed oval.

33. In the ER Toolbox, **click** the **Attribute button** and then **click a location below the major attribute in the Canvas** to add an attribute stencil to your ERD.

34. In the Canvas, **right-click** the new **Attribute oval** and then **select Properties** from the context menu to open the Properties: ER – Attribute dialog box.

35. In the Properties: ER – Attribute dialog box, **type age** in the Name: field to rename the attribute to *age*.

36. In the Properties: ER – Attribute dialog box, **click** the **No button** for the Derived: field to toggle its value to Yes; then **click** the **OK button** to close the Properties: ER – Attribute dialog box.



Derived Field Button

37. **Repeat steps 10–11** to connect the age attribute to the Student entity.

38. **Make a screen capture** showing the **ER diagram with the Student entity and attributes**.

39. In the Dia menu bar, **select File > Save** to update the saved file on the Desktop.

Part 2: Represent Relationships in an ER diagram

Note: In this part of the lab, you will establish relationships between entities and create constraints on relationships. The number of entities participating in a relationship is called the degree (or arity) of a relationship. If the degree is 2, the relationship is referred to as a binary relationship. If the degree is 3, the relationship is called ternary. If the degree is 4 or higher, the relationship is referred to as an n-ary relationship where n is the number of entities participating in the relationship.

For a binary relationship, it is common to specify the cardinality. The cardinality indicates how many times each of the two entities can participate in the relationship. For each entity, two possibilities are considered: one or many. This means there are four possible cardinalities for a binary relationship.

- one-to-one – each entity can participate in the relationship at most once
- one-to-many – one entity can participate in the relationship at most once; the other entity can participate multiple times.
- many-to-one – similar to the previous, with the roles switched
- many-to-many – each entity can participate multiple times

In addition to representing cardinality constraints, the ER model can also represent participation constraints. Participation constraints indicate whether an entity instance must participate in a relationship or whether participation in the relationship is optional. Optional participation is more flexible than total (or mandatory) participation.

The ER diagram created in Dia at the end of Part 1 is used as the starting point for Part 2. In order to create a relationship, we need a second entity. For this purpose, we will create a minimal entity representing a course.

1. In the ER Toolbox, **click** the **Entity button** and then **click a location to the right of the Student Entity in the Canvas** to add an entity stencil to your ERD diagram.

2. In the Canvas, **triple-click** the **Entity rectangle** to open the Properties: ER – Entity dialog box.

3. In the Properties: ER – Entity dialog box, **type Course** in the Name: field; then **click** the **OK button** to rename the entity to *Course* and close the Properties: ER – Entity dialog box.

Note: In the next steps, you will create a courseID identifier for the Course entity.

4. In the ER Toolbox, **click** the **Attribute button** and then **click a location to the right of the Course Entity in the Canvas** to add an attribute stencil to your ERD.

5. In the Canvas, **triple-click** the **Attribute oval** to open the Properties: ER – Attribute dialog box.

6. In the Properties: ER – Attribute dialog box, **type courseID** in the Name: field to rename the attribute stencil to *courseID*.

7. In the Properties: ER – Attribute dialog box, **click** the **No button** of the Key: field to toggle its value it to Yes; then **click** the **OK button** to close the Properties: ER – Attribute dialog box.

8. In the ER Toolbox, **click** the **Participation button** and then **click** the **courseID attribute stencil** in the drawing pane to add a connecting line that originates with the courseID attribute stencil.

9. In the Canvas, **click and drag** the **green end of the connecting line** to the Course entity to complete the connection between the courseID attribute and the Course entity.

Note: To the right of the Student ERD, you should now see a rectangle titled Course, an oval with an underlined title of courseID, and a line connecting the two shapes. This graph will be called the Course ERD. We want to create a relationship indicating that students can enroll in courses. This will be a

binary relationship between the Student entity and the Course entity. Recall that relationships are represented by a diamond shape in the ER model.

In the next steps, you will add a relationship between the two entities, Student and Course.

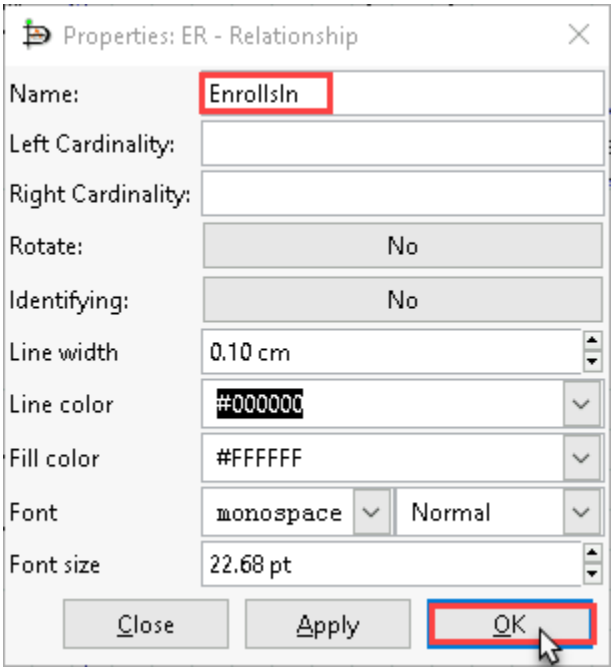
10. In the ER Toolbox, **click** the **Relationship button** and then **click a location inbetween the Student Entity and the Course Entity on the Canvas** to add a relationship stencil to your ERD.

Relationship Button

Note: For the purposes of this lab, it will be helpful to arrange the Student entity, Course entity, and EnrollsIn relationship along the same invisible horizontal line. This arrangement will be helpful in maintaining a clean visual graph when you create their connections in later steps.

11. In the Canvas, **triple-click** the **Relationship diamond** and then **select Properties** from the context menu to open the Properties: ER – Relationship dialog box.

12. In the Properties: ER – Relationship dialog box, **type EnrollsIn** in the Name: field; then **click** the **OK button** to rename the relationship to *EnrollsIn* and close the Properties: ER – Relationship dialog box.



Rename the Relationship

Note: In the naming convention for ERDs, it is customary to capitalize the names of relationships between entities.

13. In the ER Toolbox, click the **Participation button** and then **click** the **left corner** of the **EnrollsIn relationship** in the drawing pane to add a connecting line that originates with the left corner of the EnrollsIn entity.

14. In the Canvas, **click and drag** the **green end of the connecting line to the right of the Student entity** to complete the connection between the EnrollsIn relationship and the Student entity.

15. In the ER Toolbox, click the **Participation button** and then **click** the **right corner** of the **EnrollsIn relationship** in the drawing pane to add a connecting line that originates with the right corner of the EnrollsIn relationship.

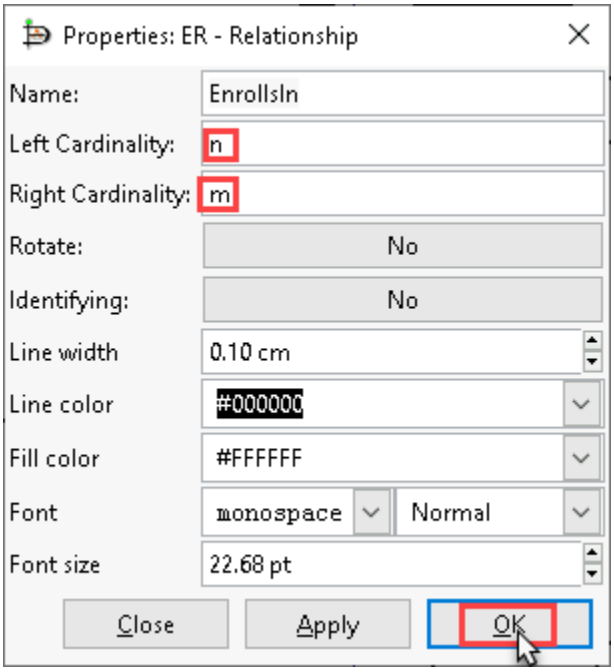
16. In the Canvas, **click and drag** the **green end of the connecting line to the left of the Course entity** to complete the connection between the EnrollsIn relationship and the Course entity.

Note: Now you need to specify the cardinality of the relationship. In an ERD, one way the cardinality can be represented is by labeling the lines on each side of the relationship with 1 to represent “one” or a letter (typically m or n) to represent “many.” For example, consider a special case in which students are working full time and are allowed to enroll in only one course at a time while working. Then the cardinality between students and courses would be many-to-one. Each student can enroll in at most one course, but each course can have many students enrolled. This type of relationship could be drawn as shown.

Many-to-one Relationship

The application we are capturing the requirements for allows a student to take multiple courses, and each course can have multiple students. Therefore, the relationship is a many-to-many relationship. In the next steps, you will add many-to-many (or m:n) cardinality to the EnrollsIn relationship in your diagram.

17. In the drawing pane, **triple-click** the **EnrollsIn relationship stencil** to open the Properties: ER – Relationship dialog box; then **type the following values** and **click** the **OK button** to close the Properties: ER – Relationship dialog box.
    Left Cardinality: **n**
    Right Cardinality: **m**



Set the Cardinality

Note: Relationships, like entities, can also have attributes. For example, a grade attribute is associated with an enrollment instance (the association between a student and a course). It would not be appropriate to associate the grade directly with the student because a student can have different grades for different courses. It also would not be appropriate to associate the grade directly with the course because many students take a course and may not all earn the same grade.

In the next steps, you will add a single-valued grade attribute to the EnrollsIn relationship.

18. In the ER Toolbox, **click** the **Attribute button** and then **click a location below the EnrollsIn Relationship in the Canvas** to add an attribute stencil to your ERD.

19. In the Canvas, **triple-click** the new **Attribute oval** to open the Properties: ER – Attribute dialog box.

20. In the Properties: ER – Attribute dialog box, **type grade** in the Name: field to rename the attribute to *grade*.

21. In the ER Toolbox, click the **Participation button** and then **click** the **bottom corner** of the

EnrollsIn relationship stencil in the drawing pane to add a connecting line that originates with the bottom corner of the EnrollsIn relationship.

22. In the Canvas, **click and drag** the **green end of the connecting line to the grade attribute** to complete the connection between the EnrollsIn relationship and the grade attribute.

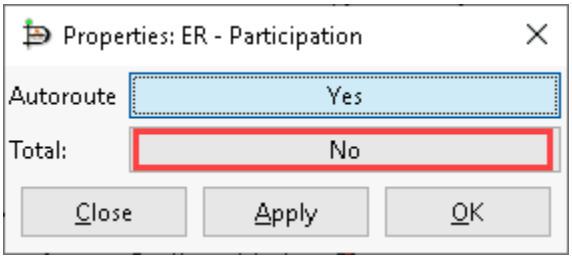
Note: Now consider the participation for the EnrollsIn relationship. Optional participation is the default in Dia and is represented by a single line from the entity to the relationship. Total participation is represented by a double line from the entity to the relationship. Participation constraints are indicated on the same side as the entity that they refer to.

In your ER diagram, the participation constraints so far are optional. That means that the ERD represents a situation in which a student does not have to be enrolled in a course and a course does not have to have students. The client you are working for requires all students to be enrolled in at least one course.

In the next steps, you will create a mandatory participation between the Student entity and the EnrollsIn relationship.

23. In the Canvas, **click** on the **connecting line between the EnrollsIn Relationship and the Student Entity** to select it; then **right-click it** and **select Properties** from the context menu to open the Properties: ER – Participation dialog box.

24. In the Properties: ER – Participation dialog box, **click** the **No button** for the Total: field to toggle its value to Yes; then **click** the **OK button** to close the Properties: ER – Participation dialog box.



Total Field Button

25. **Make a screen capture** showing the **ER diagram with the EnrollsIn relationship, Student and Course entities, and attributes for all three**.

Note: Determining the minimum and maximum number of entities that one entity instance maps to via a relationship can be confusing. It is important to remember that one entity instance is considered at a time. The numbers associated with the relationship are a maximum value, and they should be read across the relationship. For example, in the diagram shown below, one student has at most (a maximum) of 1 faculty advisor via the hasAdvisor relationship. This is determined by the 1 that appears across the relationship from Student to Faculty. A faculty member has at most m (many) student advisees, determined by reading across the hasAdvisor relationship from Faculty to Student.

Many-to-one Relationship

The minimum cardinality is determined by the participation constraints. This information is given by the line connecting the entity to the relationship. Since there is a double line from Student to hasAdvisor, a student must have a faculty advisor. Therefore, the minimum cardinality is 1. With a minimum of 1 and a maximum of 1, you can conclude that a Student instance must be associated with exactly one Faculty instance via the hasAdvisor relationship. Since the participation is optional for Faculty in hasAdvisor, there may be faculty members who do not advise any students. So, a Faculty instance maps to 0 or many (m) instances of Student via the hasAdvisor relationship.

26. In the Dia menu bar, **select File > Save** to update the saved file on the Desktop.

Part 3: Represent Weak Entities in an ER diagram

Note: If the instances of an entity can be uniquely identified by an attribute or attributes of the entity, then the entity is termed a strong entity. If an entity has a primary key attribute, it is a strong entity. The Student and Course entities are examples of strong entities. Some entities depend on other entities for unique identification and are only partially identified by one or more of their own attributes. This kind of entity is called a weak entity. A weak entity does not have a primary key, but it does have

a weak key (also known as a partial key or a discriminator).

Since a weak entity relies on another entity to provide additional information to uniquely identify its instances, it must be related to another entity via a relationship. This relationship is called its identifying relationship. Identifying relationships are denoted by a double diamond. They usually have 1:n cardinality and total participation on the weak entity side.

The university divides classes into sections. Each section is designated by letters such as A, B, C, and so on. For a given course, the section names are unique. However, sections are not unique by themselves. More than one course can have a section A, for example.

In this part of the lab, you will create a weak entity to represent sections, with the section letter acting as a partial key and a relationship to the Course entity acting as the identifying relationship.

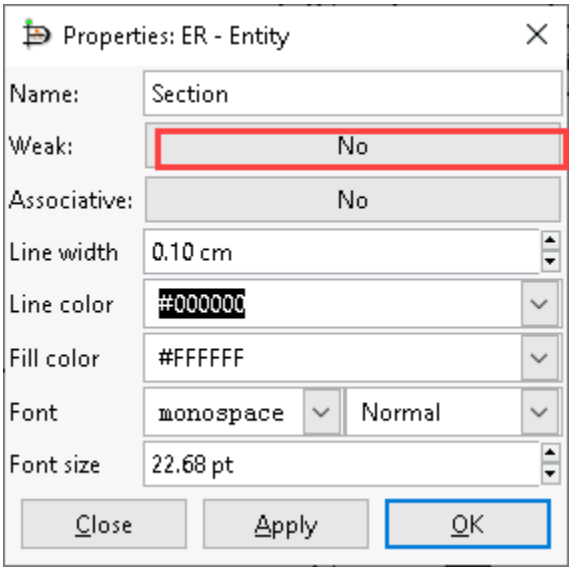
1. In the ER Toolbox, **click** the **Entity button** and then **click a location below the Course entity stencil on the Canvas** to add an entity stencil to your ERD.

2. In the Canvas, **triple-click** the **Entity rectangle** to open the Properties: ER – Entity dialog box.

3. In the Properties: ER – Entity dialog box, **type Section** in the Name: field to rename the entity to *Section*.

Note: A weak entity is denoted as a double rectangle.

4. In the Properties: ER – Entity dialog box, **click** the **No button** for the Weak: field to toggle its value to Yes; then **click** the **OK button** to save your changes.



Weak Field Button

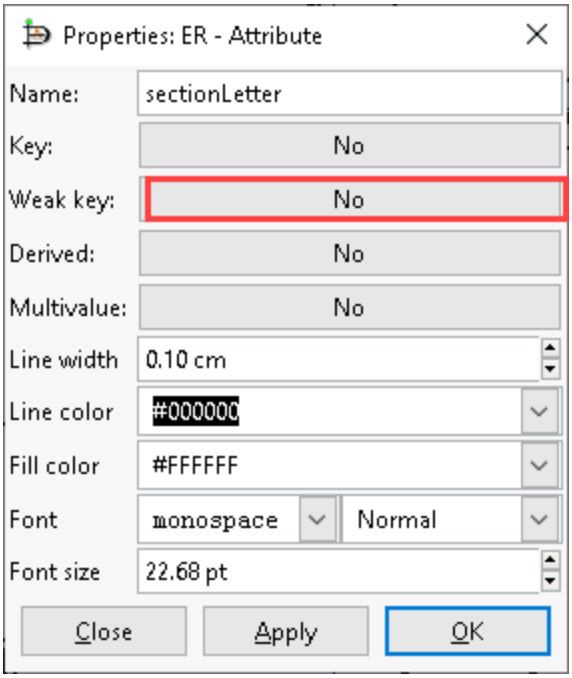
Note: You should see that the Section entity rectangular stencil now has a double border. A weak key is often indicated by a dashed underline.

5. In the ER Toolbox, **click** the **Attribute button** and then **click a location to the right of the Section Entity in the Canvas** to add an attribute stencil to your ERD.

6. In the Canvas, **triple-click** the **Attribute oval** to open the Properties: ER – Attribute dialog box.

7. In the Properties: ER – Attribute dialog box, **type sectionLetter** in the Name: field to rename the attribute to *sectionLetter*.

8. In the Properties: ER – Attribute dialog box, **click** the **No button** for the Weak key: field to toggle its value to Yes; then **click** the **OK button** to close the Properties: ER – Attribute dialog box.



Weak Key Field Button

9. In the ER Toolbox, **click** the **Participation button** and then **click** the **sectionLetter attribute stencil** in the drawing pane to add a connecting line that originates with the sectionLetter attribute stencil.

10. In the Canvas, **click and drag** the **green end of the connecting line** to the Section entity to complete the connection between the sectionLetter attribute and the Section entity.

Note: You should see that sectionLetter is underlined with a dashed line inside its oval stencil. This graphically indicates that the sectionLetter attribute is a weak key.

Each section instance must be associated with exactly one course to uniquely identify that section. Section A of CSE 385 is a different instance than Section A of CSE 485, but CSE 385 may have many instances of sections (A, B, C, D, and so on).

In the next steps, you will create the identifying relationship for the section entity.

11. In the ER Toolbox, **click** the **Relationship button** and then **click a location between the Course Entity and the Section Entity stencils on the Canvas** to add a relationship stencil to your ERD.

Note: For a clean graph, you may need to move the Section Entity and sectionLetter attribute stencils down to make space for the relationship stencil.

12. If necessary, **click and drag** the **Section entity, the sectionLetter attribute, and/or the Relationship stencils** with your mouse to arrange them so that the Relationship entity is situated vertically between the Course and Section entity stencils.

13. In the Canvas, **triple-click** the **Relationship diamond** to open the Properties: ER – Relationship dialog box.

14. In the Properties: ER – Relationship dialog box, **type the following values** and **click** the **OK button** to close the Properties: ER – Relationship dialog box.

    Name: **HasSection**
    Left Cardinality: **1**
    Right Cardinality: **n**
    Identifying: **Yes**

Note: You should see that the HasSection Relationship stencil has a double border to indicate that it involves a weak entity. You should also see the number 1 adjacent to its left corner and the letter n adjacent to its right corner. These represent the types of participation that the course and section entities have.

In the next steps, you will complete the participation between the Course entity, Section entity, and HasSection relationship.

15. In the ER Toolbox, click the **Participation button** and then **click** the **top corner** of the **HasSection relationship** in the drawing pane to add a connecting line that originates with the top corner of the HasSection relationship.

16. In the Canvas, **click and drag** the **green end of the connecting line to the bottom of the**

Course entity to complete the connection between the HasSection relationship and the Course entity.

17. In the ER Toolbox, click the **Participation button** and then **click** the **bottom corner** of the **HasSection relationship** in the drawing pane to add a connecting line that originates with the bottom corner of the HasSection relationship.

18. In the Canvas, **click and drag** the **green end of the connecting line to the top of the Section entity** to complete the connection between the HasSection relationship and the Section entity.

19. In the Canvas, **click** on the **connecting line between the HasSection Relationship and the Section Entity** to select it; then **right-click it** and **select Properties** from the context menu to open the Properties: ER – Participation dialog box.

20. In the Properties: ER – Participation dialog box, **click** the **No button** for the Total: field to toggle its value to Yes; then **click** the **OK button** to close the Properties: ER – Participation dialog box.

Note: You should notice that the Left Cardinality (1) and Right Cardinality (n) values do not align with the connecting lines between the Course and Section.

21. In the Canvas, **triple-click** the **Relationship diamond** to open the Properties: ER – Relationship dialog box.

22. In the Properties: ER – Relationship dialog box, **click** the **No button** of the Rotate: field to toggle its value to Yes; then **click** the **OK button** to close the Properties: ER – Relationship dialog box.

Note: You should notice that the Left Cardinality (1) and Right Cardinality (n) values now align with the connecting lines between the Course and Section. You should also see that the Left Cardinality value (1) corresponds with the connection between the Course entity and HasSection relationship, while the Right Cardinality value (n) corresponds with the connection between the Section entity and the HasSection relationship. This placement indicates that one Course can have n (many) Sections.

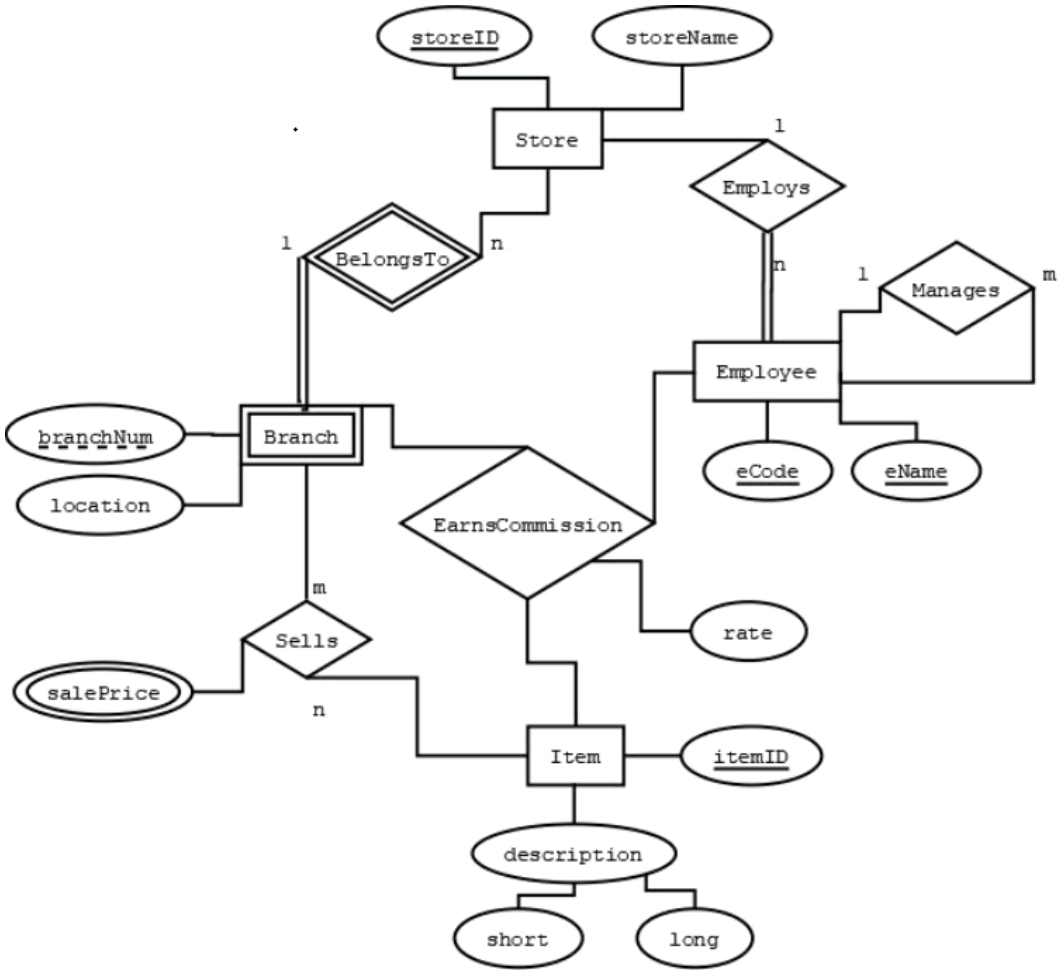
23. **Make a screen capture** showing **the entire ERD**.

Challenge and Analysis

Note: The following scenario is provided to allow independent, unguided work, similar to what you will encounter in a real situation.

Part 1: Analyze an ER diagram

Another member of your design team has been working with a different client to create the conceptual design for a store. The designer has asked you to review their work so far to check for any issues. Here is the current ERD for the application (***also available on vWorkstation at StoreConceptualDesign.jpg***):



ER Diagram

Here are a few hints for your work:

- Strong entities are represented by rectangles and must have primary keys.
- A weak entity is represented by a double rectangle and must have an identifying relationship (double diamond) with an owner entity, total participation of the weak entity in the identifying relationship, usually 1:n cardinality, and a weak key (dashed underline).
- If more than one attribute is underlined for an entity, then that entity has a compound or composite key.
- A multi-valued attribute is represented by a double oval.
- A composite attribute is represented by two levels of attributes.
- Relationships are represented by diamonds.
- Binary relationships have exactly two entities participating in the relationship.
- A relationship with three entities participating in it is termed a ternary relationship.
- Total participation is represented by a double line from the entity to the relationship.

Here is a list of what your coworker would like:

1. **Document** the names of all of the **strong entities and their primary key attributes** in the diagram.
2. **Document** the names of all of the **binary relationships and their cardinalities** in the diagram.
3. **Document** the names of the **ternary relationships** in the diagram.
4. **Document** the names of **entities** and **relationships** where the entity has total participation in the relationship.

5. **Document** the names of the **weak entity**, its **identifying relationship**, its **owner entity**, and its **weak key**.

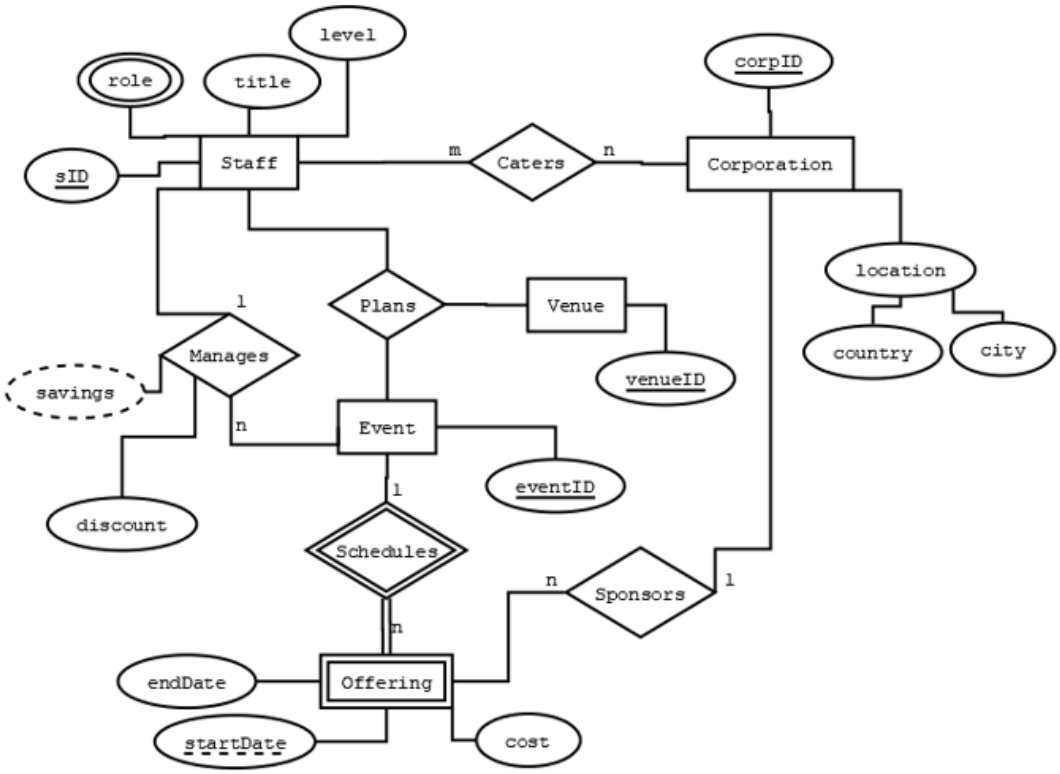
6. **Document** the names of **multi-valued attributes** and the **entity or relationship** they belong to.

7. **Document** the names of **composite attributes** and the **entity or relationship** they belong to.

Part 2: Create an ER diagram

As a newly hired database designer, you are preparing to create a database for a client. Your predecessor in the position was promoted to a DBA role in another part of the organization, but she left you some preliminary work that she had completed. She worked with the client to create and validate an ER diagram that captures their requirements. Your manager would like for you to (1) recreate the diagram using a drawing tool (Dia) and (2) explain aspects of the diagram to demonstrate that you understand the client's requirements.

1. **Create the ER diagram** for a catering application given here in **Dia:**



ER Diagram

2. **Take a screen capture** of your **catering application ER diagram final result** in the Dia Canvas.

Part 3: Analyze an ER diagram

Using the ER diagram for the catering application, give your answers to questions about the entities, attributes, and relationships depicted in the diagram. You will be asked to discuss entities, attributes (primary key, single-valued, multi-valued, composite), relationships (degree, cardinality, participation), and weak entities (weak key, identifying relationship, owner entity, total participation) in the ER diagram.

1. **Document** which **attribute or attributes** uniquely identifies the entities (1) Staff, (2) Corporation, (3) Event, and (4) Venue.

2. **Document** all of the **ER diagram elements** associated with the weak entity Offering.

3. **Document** the **combination of attributes** needed to uniquely identify an instance of Offering (its weak key and its owner entity's primary key).

4. Document the minimum (based on participation constraints) number of Staff that manage an Event, and the maximum (based on cardinality constraints) number of Events that Staff can manage.

5. Document the minimum (based on participation constraints) number of Offering instances related to an Event through Schedules, and the maximum (based on cardinality constraints) number of Events related to each Offering instance.

6. **Document the minimum (based on participation constraints) number of Corporation instances related to Staff through Caters, and the maximum (based on cardinality constraints) number of Staff that can cater each Corporation instance.**

7. **Document whether a value is associated with each attribute in the composite attribute** shown in the diagram.

8. **Document whether an instance of Staff can have 0 or more values for** (1) roles, (2) title, and (3) sID.

9. **Document** what the **dashed oval** means for the attribute savings of the Manages relationship.